Circular economy strategies (CE)

Sand substitution
More intensive use
End-of-life recycling
Clinker substitution
Lifetime extension
Component reuse

IMAGINE-Concrete

Extraction
gravel
sand
limestone
Manufacturing
Construction
In-use stock
End-of-life

Mining area simulation

Projecting future aggregate extraction demand

Existing building material stocks

Growth pattern (BAU)

Per-capital stock in grid cells converge to the median of cells with similar GDP per capita and population density

Shared Socioeconomic Pathway 2 (SSP2)

Spatial profiles of per-capita in-use stock

Future population distribution

Future GDP distribution

Future distribution of building material stocks

Future growth factors by 2050

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Extraction demand for stock turnover

Extraction demand for stock growth

Downscaling extraction demand

Grid-level extraction demand for replacing retired stock

Grid-level extraction demand for stock growth

Grid-level extraction demand

Future distribution of building stock growth

Mining area simulation

Spatial modeling of aggregate mining suitability

Quarry presence locations (from OpenStreetMap)

Distance constraint layers

Presence grid cells

Pseudo-absence grid cells

Environmental and socio-economic predictors

Random Forest classifier

probability output

Predicted mining suitability

Fitting the extraction-area relationship

Selected quarry samples

Quarry pit polygons (Manually Delineated in Google Earth Pro)

Estimated disturbed area

Estimated cumulative extraction

Fitted extraction-area relationship

Estimating accessible aggregate reserve

Predicted mining suitability

Land-use & protection masks

Screen by threshold

Lithology-based volumetric yield

Available land

Accessible reserve (1 km)

Aggregated reserve (50 km)

Mining area estimation

Available land

Remaining reserve

Mining suitability

Sub-cell allocation and estimation

Grid-level mining area expansion

Extraction demand-reserve allocation

Grid-level extraction

Demand-reserve matching

Biodiversity risk assessment

Species range polygons

Map of terrestrial habitat types

Digital elevation model

Area of habitat (AOH)

Rarity-weighted richness (RWR)

Mining-biodiversity overlap

Conservation-oriented (CON)

Siting toward low-RWR areas

Legend

Model
tabular data
Point/Polygon data
Spatial resolution
raster data
1-km
50-km